



Modal Analysis of Transmission Housing Using Finite Element Modeling for Various Assembly Conditions

Starvin Michael Stanis Anna University

Venugopal Nair and Ganesh KC R&D, AVTEC Limited, Hosur - 635114

Adharsh M and Ajith R Anna University

Citation: Stanis, S.M., Nair, V., Ganesh, K.C., Adharsh, M. et al., "Modal Analysis of Transmission Housing Using Finite Element Modeling for Various Assembly Conditions," SAE Technical Paper 2020-28-0367, 2020, doi:10.4271/2020-28-0367.

Abstract

In recent years the solution of the dynamic behavior of a structure by the application of the Finite Element Method has become more popular. Modal analysis is the process of determining the inherent dynamic characteristics of a system. The solver used in this analysis will give the Eigen values and Eigen vectors which represent the natural frequencies and corresponding mode shapes by solving the Eigen system. The extracted natural frequencies and mode shapes define the dynamic characteristics of the system. Free-free and constrained modal analyses are the two major types, which explains the dynamic nature of the system with and without assigned boundary conditions respectively. In this research project, the main aim is to arrive an appropriate modal analysis procedure and setting up a guideline to execute the modal analysis of transmission housing assembly

using numerical analysis based finite element modeling by performing both free-free and constrained modal analysis for various assembly conditions. Modal analysis of transmission housing is a challenging task because of the complexity in its structure and the presence of various internal components. In free-free modal analysis the entire assembled model is analyzed with assumptions like surface to surface interactions and one dimensional beam elements separately, in order to achieve the bolted connections. Further in constrained modal analysis one more assumption is added that the mass of internal components and oil. In all the cases, the natural frequencies and mode shapes are extracted, documented and compared to investigate the influence of the various assembly conditions and ensure that transmission housing will not meet the resonant condition in its operating range.

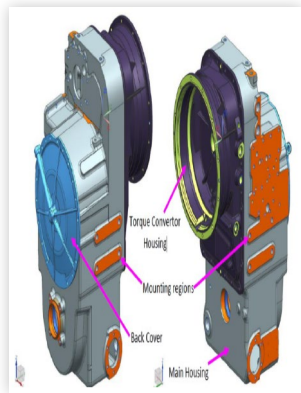
Introduction

Modal analysis is the study of the dynamic properties of systems in the frequency domain to determine the natural mode shapes and frequencies during the vibrational excitation. Free-Free and Constrained modal analysis are the general modal analysis techniques in FEM approach. The main purpose of the Free - Free Modal analysis is to understand how the body behaves when it is unconstrained. Hence first six frequencies will be almost zero from seventh onwards natural frequency value will be obtained. In case of constrained modal analysis, the geometry features are fixed in a place or in particular directions and probably the first Eigen frequency will give the natural frequency value.

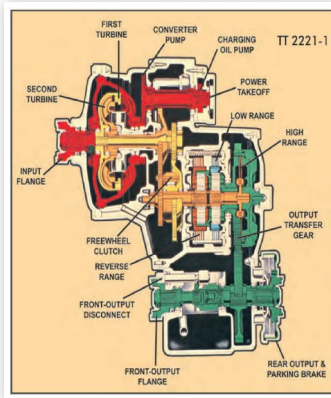
The main purpose of the modal analysis on the transmission housing is to extract natural frequencies and mode shapes of the system under different conditions. The condition at which the natural frequency closer to the excitation frequency is known as resonance condition and in this condition the whole system will resonate for higher amplitude. Hence the results of a modal analysis help to determine whether a model

requires more or less damping to prevent this condition. The mode shape always describes the time to time curvature of the vibrations and it depends on the shape and the boundary conditions of the surface.

The Eigen-values and Eigen-vectors are the direct results of an Eigen solver and these are the parts of the stiffness matrix. An Eigen-value and its associated Eigenvector can be interpreted as a strain energy and corresponding displacement mode, respectively. On the basis of this physical meaning, an Eigenvector can be graphically visualized as a displacement mode which results in the strain energy relevant to the associated Eigen value. The set of an Eigen value and associated Eigenvector is termed "Eigen mode". There are as many Eigen modes as the number of free degrees of freedom in the model, which may consist of a single element, a number of selected elements, or the whole system. The model may or may not include the boundary constraints, which are reflected in Eigen modes. The integration rules as well as the element properties assigned to the model are also reflected in the displayed Eigen modes.

FIGURE 1 Housing Assembly

© SAE International and SAE India.

FIGURE 2 Internal components

© SAE International and SAE India.

In this project, the transmission housing assembly subjected to the modal analysis consists of torque converter housing, main housing and back cover. All three parts are connected with bolted joints in the final assembly as shown in [figure 1](#).

The internal arrangements of components are shown in [figure 2](#), which distributes mass across the bearing mounts within the assembly along with specific quantity of transmission fluid.

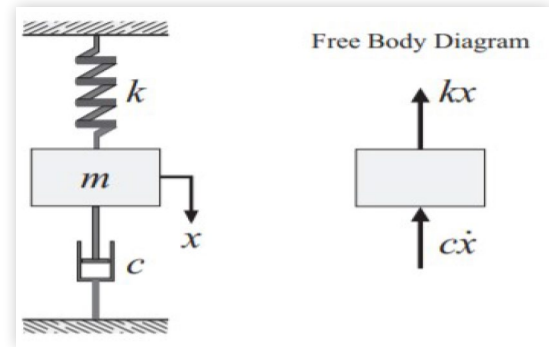
The following assumptions are considered in the entire analysis process.

- With and without bolt connections between housings
- With and without mass of internal components

For each trial the corresponding constraints are used as per the assumption considered to obtain the relevant solutions.

The natural frequency can be extracted from a vibratory system when it will be in equilibrium. This is in accordance with D'Alembert's principle that the sum of all external forces and inertia forces on a body in equilibrium will be zero.

Consider a body of mass (m) supported by a spring of stiffness (k) and attached to a dash pot, whose resistance may be considered proportional to the relative velocity. Let the mass (m) be given a downward (i.e., positive) displacement from the equilibrium position and released. At some time (t) the mass will be at a distance (x) from the equilibrium position.

FIGURE 3 Spring-Mass-Damper system

© SAE International and SAE India.

The spring force ($k \cdot x$) acting on the body will tend to restore it to its equilibrium position and the damper force tending to oppose motion will be $\left(c \frac{dx}{dt}\right)$ where c is the viscous damping coefficient.

By summing the forces in the vertical direction and assuming motion about the static equilibrium position (refer to the free body diagram),

$$\Sigma F = ma, \text{ which yields} \quad (1)$$

$$-c\dot{x} - kx = m\ddot{x} \quad (2)$$

Where, $\dot{x}$ and $\ddot{x}$ are the first and second order derivative of the displacement (x) respectively. This equation can be written as:

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (3)$$

In case of forced vibration, the value of external force (F_0) will also consider:

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin(\omega t) \quad (4)$$

By dividing the [Equation \(3\)](#) with mass (m):

$$\ddot{x} + \left(\frac{c}{m}\right)\dot{x} + \left(\frac{k}{m}\right)x = \left(\frac{F_0}{m}\right)\sin(\omega t) \quad (5)$$

With the following condition:

$$\omega^2 = (k/m) \text{ and } 2\zeta\omega = (c/m) \quad (6)$$

Where, ω is the natural frequency and ζ is the damping ratio for the given spring-mass-damper system. The entire equation can be rewritten as:

$$\ddot{x} + 2\zeta\omega\dot{x} + \omega^2 x = \left(\frac{F_0}{m}\right)\sin(\omega t) \quad (7)$$

Generally the modal analysis would do for free un-damped vibrations; where, the damping coefficient and the external forces will be zero. These are the basic equations involved during the determination of the dynamic behavior of the structure.

Literature Review

Teik C. Lim, Raj. Singh and James J. Zakrajsek [1] conducted a dynamic finite element analysis of real gear housing without

the rotating components (gears, shafts, and bearings). Both rigid and flexible mounting conditions are considered in this analysis. The flexible support simulates the realistic mounting condition on a rotor craft, and the rigid one is analyzed for comparison purposes. The effect of gear housing stiffeners is also evaluated. Studies show that the housing plate stiffeners and rigid mounts tend to increase most of the natural frequencies, the lower ones being affected the most. **Dogan [2]** varied the transmission parameters like main center distance, axial clearance and backlash by experimental analysis in test stand trials in order to reduce undesirable noise of rattle and clattering in transmission gear box due to torsional vibration. **Y. L. Xu, and W. S. Zhang [3]** studied 'Modal analysis and seismic response of steel frames with connection dampers'. Effects of connection stiffness and rotational damper on natural frequency and modal-damping ratio were enquired. The parametric studies on the example frame with and without the connection dampers showed that there was an optimal damper damping coefficient by which the modal-damping ratio of the frame can be considerably increased and the seismic responses, including both lateral displacements and internal forces, can be significantly reduced. **Hong HeeYooa, Jung Eun Chob, and JintaiChungc [4]** studied Modal analysis of cantilever beams undergoing rotation and its shape and shape optimization. In this study, the geometric shapes that satisfy the modal characteristic requirements such as maximum or minimum slope natural frequency are obtained through an optimization procedure. **Chavan, Mahale and Thakur [5]** presented a research work of Modal analysis of Power Take off Gearbox to understand the generation of the vibration in PTO gearbox and provide recommendations to reduce vibrations by 15-20% of current levels. **Abouel-Seoud and Abdullah [6]** developed a concept of a method for predicting the contribution to exterior noise radiated by an original vehicle gear box structure using measured vibration response. **Kumar and Patil [7]** done a research work signifies the material-based effect on natural frequencies and mode shapes and weight of housing was calculated for three different materials. Finally concluded with that a new alloy steel AISI 4130 having chromium and molybdenum as chief elements for strength enhancement is recommended for future application of transmission gearbox housing based on design and vibration index. **Rucha, Kulkarni and Purandare [8]** presented a paper which explains simple steps of the modal analysis simulation using ANSYS®. The Cantilever beam is taken for study and frequencies and mode shapes are obtained in the graphical environment of ANSYS. Further theoretical natural frequencies are obtained by hand calculations and compared with that obtained from FEM. **Vijay Kumar, Shivaraju and Srikanth [9]** investigated about vibration analysis for gearbox casing using Finite Element Analysis (FEA) using ANSYS software. This analysis is to find the natural frequency and harmonic frequency response of gearbox casing in order to prevent resonance for gearbox casing. From the result, this analysis showed the range of the frequency that is suitable for gearbox casing which can prevent maximum amplitude. **Gopali S Lamani and Basavaraddi [10]** conducted a modal analysis on Power Take Off (PTO) for two materials Aluminum alloy and Grey CI using both analytical and theoretical method. Modal analysis

is used to measure the natural frequency of the system and different mode shape patterns are observed for both Aluminum alloy and Grey CI. Finally concluded that Aluminum is having lighter weight than Grey CI and which is recommended in the manufacturing of PTO gearbox housing. **Vidhya and Srinivas [11]** attempts to present numerical solution for the vibration analysis of a Flux Reversal Generator (FRG). Natural frequency and mode shapes of the generator have been calculated by mean of the FE analysis. From the results, it could be seen that the bending mode frequency is 3848.85 Hz, which causes high vibration and noise. Harmonic response could be performed in the frequency domain based on the calculated frequencies. The large displacement can occur when centrifugal force frequencies are near the natural frequency of modes 1 and 2. **Peter Weis, LubosKucera, Peter Pechac and Martin Mocilan [12]** presented an examination of the modal characteristic of gearbox housing with applied load. The first step involved solution of static analysis with load in form of bearing reactions. Subsequently modal analysis was performed to obtain natural frequencies and modal shapes. Finite element analyses were performed with ANSYS Workbench software. Natural frequencies were computed and concluded that the tendency to oscillate occurs mainly in the area of flange.

Numerical Analysis

The CAD model for the entire transmission housing is imported into Abaqus which consists of inserts, main housing, torque convertor housing and back cover. The material properties used in this analysis for various parts are listed below,

TABLE 1 Material properties

Part	Material	ρ (tonne/mm ³)	Poisson ratio	E (Mpa)
Main housing	Grey CI (G2500)	7.13 E ⁻⁹	.29	180000
T.C Housing	Al AC4B	2.77 E ⁻⁹	.33	73000
Back cover	A360 Al Alloy	2.63 E ⁻⁹	.3	70000
Inserts	Steel	7.8E ⁻⁹	.3	206000

© SAE International and SAE India.

Using multi point constraint, coupling and tie, the required constraint conditions are constructed, shafts which are carrying internal components are modeled using 1D beam elements and the mass of internal component are provided using inertia. In step module linear perturbation procedure was selected and Lanczos eigen solver is opted to extract the natural frequencies of the system. The Lanczos algorithm is one of the most popular methods for obtaining low energy eigen vectors and eigen values of sparse matrices. For the Lanczos method you need to provide the maximum frequency of interest or the number of eigenvalues required.

The loads or constraints that represent the effect of the surrounding environment on the model are generally known as boundary conditions. In actual case the entire transmission housing is fixed on the chassis of the vehicle and the free end

TABLE 2 Mass details

SL.NO	COMPONENT	MASS (ton)
1	Torque Converter	0.045
2	Gear train	0.035
3	Output shaft	0.035
4	PTO Gears per bore	0.012
5	Oil mass at CG	0.030

© SAE International and SAE India.

of the torque converter is fixed with the engine block. By this bolted connections, the fixed parts neither translate nor rotate. In order to achieve this condition, ($U_1 = U_2 = U_3 = UR_1 = UR_2 = UR_3 = 0$) is used as a boundary condition in fixed regions. Meshing is a very important process in FEA approach because it actually makes the problem solvable by Finite Element Method. FEA reduces the degrees of freedom from infinite to finite with the help of discretization or meshing (nodes and elements). The model was meshed using a 10 node tetrahedral element. Mesh sensitivity analysis was carried out and at the end of the process mesh size of 15mm was selected. In this project modal analysis, various assumptions are made. The various assumed conditions during analysis are listed below,

TABLE 3 Assembly Conditions

Case	Type	Assumption
1	Free-Free Modal Analysis	With surface contact conditions
2	Free-Free Modal Analysis	1D rigid beam element modeled instead of 3D bolted connections.
3	Constrained Modal Analysis	1D rigid element modeled instead of 3D bolted connections and without the mass of internal components& oil.
4	Constrained Modal Analysis	1D rigid beam element modeled instead of 3D bolted connections, 3D Shafts and with mass of internal components& oil.

© SAE International and SAE India.

After completing modal analysis of the various assembly conditions, the natural frequencies are listed and the mode shapes are animated in the visualization module. All the results are documented.

Results and Discussion

The corresponding natural frequencies are tabulated below for the various assembly conditions.

In case 1, the transmission housing is subjected to free-free analysis with surface-surface contact. The results of case1 explain how the system behaves when it is completely unconstrained. In this analysis the natural frequencies are obtained from the seventh mode. In this analysis the natural frequencies obtained from 546.20 to 795.59 Hz.

TABLE 4 Natural Frequencies of various assembly conditions.

MODE	EIGEN FREQUENCY FOR FREE-FREE MODAL ANALYSIS (Cycles/time)		EIGEN FREQUENCY FOR CONSTRAINED MODAL ANALYSIS (Cycles/time)	
	Case 1	Case 2	Case 3	Case 4
1	0.00	0.00	479.51	342.23
2	0.00	0.00	535.31	367.59
3	0.00	0.00	712.23	457.70
4	0.00	0.00	764.29	487.34
5	0.00	0.00	798.55	523.25
6	0.00	0.00	910.10	567.74
7	546.20	538.20	964.73	632.72
8	676.88	670.67	974.88	699.25
9	732.55	726.81	996.48	773.36
10	795.59	793.37	1047.2	851.65

© SAE International and SAE India.

In case 2, the natural frequencies obtained from 538.20 to 793.37 Hz. This frequency is little low when compared to the results of case1 and the reduction in the natural frequency for the tenth mode of case2 is 0.27% compared with case1. This deviation is occurred due to the 1D beam element (It will enable some displacement constraints on the subjected regions), which replaces 3D shafts in this analysis.

In case 3, the only difference from free-free modal analysis (case1&case2) is that the boundary conditions applied on the fixed regions on the housing (i.e. the regions mounted on the chassis). The non-zero natural frequencies obtained from the first mode and it varies from 479.51 to 1047.2 Hz. In this case the natural frequencies are extracted for mode number '10' is high, when compared to case2. The percentage of increment in the natural frequency for the tenth mode of case3 is 24 %, because of the change in displacement matrix due to boundary conditions.

In case 4, the mass of the internal components are considered. The values of natural frequencies extracted are less than previous cases, in this analysis natural frequencies are obtained from 342.23 to 851.65 Hz. The percentage reduction of natural frequency for tenth mode of case4 when compared to the case3 is 18.67%. Natural frequency vary inversely proportional to the mass.

$$\omega = \sqrt{k/m} \quad (15)$$

While adding mass it reduces the value of natural frequency of constrained modal analysis (with mass of internal components& oil) compared with natural frequency of constrained modal analysis (without mass of internal components& oil).

Guidelines for Modal Analysis

- Import model.
- Select proper mesh technique for meshing of the component depends upon the requirements.

- Perform mesh sensitivity analysis and fix the element size for meshing.
- Check the mesh quality to minimize the formation of poor elements, which may cause deviation in results.
- The following material properties are mandatory to perform a modal analysis are,
 - Density
 - Modulus of elasticity
 - Poisson's ratio
- Use surface contact definition as well as 1D rigid beam element instead of bolted connections to avoid relative tangential motion of individual components, so that it will act as a single system during analysis.
- Assign mass of internal components using 1D rigid beam element and coupling constraints, if the mass of the internal components are considered.
- Assign linear perturbation procedure with frequency to define modal analysis in FEA solver.
- And select lanczos Eigen solver to compute mode shapes and natural frequency of the system. For the Lanczos solver we need to provide the maximum frequency of interest or the number of Eigen values required.
- Then provide zero displacement constraints to the respective fixed regions which are rest on the chassis using boundary conditions.
- After completion of pre-processing, check the input data's provided is errors free.
- Run the analysis.
- In post-processing, visualize the mode shapes and check the natural frequency of the system.
- The extracted natural frequencies and the corresponding mode shapes will be documented for future reference.

Conclusion

In this research project modal analysis of transmission housing is performed for various assembly conditions. Natural frequencies for various cases are extracted and tabulated accordingly. From this tabulation it can be conclude that all the natural frequencies are far away from the operating range of the transmission housing and will not meet resonant condition. At the end of this analysis an appropriate procedure with guidelines to execute the modal analysis of transmission housing for various assembly conditions is arrived using numerical analysis based finite element modeling.

References

1. Lim, T.C., Singh, R., and Zakrajsek, J.J., "Modal Analysis of Gear Housing and Mounts," NASA Technical Memorandum, 1989.
2. Dogan, S.N., "Loose Part Vibration in Vehicle Transmissions- Gear Rattles," *Trans. J. Eng. Environ. Sci* 23, 1999.
3. Xu, Y.L., and Zhang, W.S., "Modal Analysis and Seismic Response of Steel Frames with Connection Dampers," *Engineering Structures*, 2001.
4. Hee, Y.H., Eun, C.J., and Jintai, C., "Modal Analysis and Shape Optimization of Rotating Cantilever Beams," *Journal of Sound and Vibration*, 2006.
5. Chavan, D.S., Mahale, A.K., and Thakur, A.G., "Modal Analysis of Power Take Off Gearbox," *International Journal of Emerging Technology and Advanced Engineering*, 2013.
6. Abouel-Seoud, S.S., Mohamed, E.S., Abdel-Hamid, A.A., and Abdullah, A.S., "Analytical Technique for Predicting Passenger Car Gearbox Structure Noise Using Vibration Response Analysis," *Br. J. Appl. Sci. Technol*, 2013.
7. Kumar, A., and Patil, P.P., "Free Vibration and Connecting Bolt Constraint-Based FEA Analysis of Heavy Vehicle Medium Duty Transmission Gearbox Housing Made from AISI 4130 Alloy Material," *Journal of Engineering Science and Technology Review*, 2014.
8. Rucha, S., Bhakte, A.P., Kulkarni, P.S., and Purandare, "An Overview of Modal Analysis Using Finite Element Method," *IJSRD-International Journal for Scientific Research & Development*, 2014.
9. Kumar, V., and Shivaraju, S., "Vibration Analysis for Gearbox Casing Using Finite Element Analysis," *The International Journal of Engineering and Science (IJES)*, 2014.
10. SLamani, G., and Basavaraddi, S.R., "Static and Modal Analysis of Tractor Power Take off (PTO) Gearbox Housing," *International Research Journal of Engineering and Technology (IRJET)*, 2016.
11. Vidhya, B., and Srinivas, K.N., "Vibration Analysis Including Stator, Rotor, Housing and Dynamic Response Analysis of Flux Reversal Generator," *Journal of Electrical Systems and Information Technology*.
12. Weis, P., Kucera, L., Pechac, P., and Mocilan, M., "Modal Analysis of Gearbox Housing with Applied Load," in *International Scientific Conference on Sustainable, Modern and Safe Transport*, 2017.
13. Abaqus Documentation, <http://dsk.ippt.pan.pl/docs/abaqus/v6.14/index.html>.
14. <http://www.matweb.com/search/datasheet.aspx?MatGUID=f3cd25980ab24fdaa5893252cd2bc192>.
15. http://www.efunda.com/materials/alloys/aluminium/show_aluminium.cfm?ID=AA_360.0A&show_prop=all&Page_Title=A360%2E0.
16. Lim, T.C., Singh, R., and Zakrajsek, J.J., "Modal Analysis of Gear Housing and Mounts," NASA Technical Memorandum, 1989.